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AI Fake News Detection using Machine Learning and Natural Language Processing

Project Report

15th June to 30th July 2026

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AI-Based Fake News Detection Using Machine Learning and Natural Language Processing

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***Abstract*—The rapid growth of digital media and online news platforms has significantly increased the spread of fake and misleading information. Manual verification of news articles is time-consuming and cannot efficiently handle the enormous volume of information generated every day. This paper presents an Artificial Intelligence-based Fake News Detection System that utilizes Natural Language Processing (NLP) and Machine Learning techniques to automatically classify news articles as either Real or Fake.**

The proposed system employs a complete machine learning pipeline consisting of dataset preparation, text preprocessing, TF-IDF feature extraction, model training, evaluation, and deployment through an interactive Streamlit web application. The preprocessing stage includes lowercase conversion, URL removal, punctuation removal, stop-word elimination, lemmatization, and whitespace normalization to improve text quality. The cleaned news articles are transformed into numerical feature vectors using Term Frequency–Inverse Document Frequency (TF-IDF) with a vocabulary size of 5000 features.

Four supervised machine learning algorithms, namely Logistic Regression, K-Nearest Neighbors (KNN), Random Forest, and Multi-Layer Perceptron (MLP) Neural Network, are trained and evaluated using Accuracy, Precision, Recall, F1-Score, and ROC-AUC metrics. The best-performing model is automatically selected and integrated into a Streamlit-based dashboard that supports real-time prediction, batch prediction using CSV files, prediction confidence visualization, probability charts, prediction history, and interactive model performance analysis.

The proposed system provides an efficient, scalable, and user-friendly solution for fake news detection and demonstrates the practical application of Machine Learning in combating misinformation across digital platforms.

***Index Terms*—Artificial Intelligence, Machine Learning, Natural Language Processing, Fake News Detection, TF-IDF, Text Classification, Logistic Regression, Random Forest, Neural Network, Streamlit, Interactive Dashboard**

I. Introduction

The exponential growth of the Internet, online news portals, and social media platforms has transformed the way information is created, distributed, and consumed. Millions of news articles are published and shared every day through websites, blogs, and social networking platforms, allowing information to spread globally within a few seconds. While this digital transformation has improved communication and accessibility, it has also contributed to the rapid spread of misinformation and fake news.

Fake news refers to intentionally fabricated, misleading, or manipulated information presented as legitimate news with the purpose of influencing public opinion, generating financial benefits, or creating political and social instability. During critical events such as elections, pandemics, natural disasters, and international conflicts, false information spreads much faster than verified news, making manual verification increasingly difficult. The widespread circulation of misinformation can affect public trust, healthcare awareness, financial markets, democratic processes, and social harmony.

Traditional fact-checking methods rely heavily on human experts and trusted organizations. Although these methods provide reliable verification, they require considerable time and effort and cannot efficiently process the massive volume of digital news generated daily. Therefore, intelligent automated systems capable of identifying fake news have become an important area of research.

Artificial Intelligence (AI), Machine Learning (ML), and Natural Language Processing (NLP) provide effective solutions for automatic fake news detection. NLP techniques enable computers to understand textual information by extracting meaningful linguistic patterns, while Machine Learning algorithms learn classification patterns from previously labelled news articles. These technologies significantly reduce manual effort and improve the speed and consistency of fake news identification.

This work presents an AI-Based Fake News Detection System developed using Machine Learning and Natural Language Processing techniques. The proposed system employs a complete end-to-end machine learning pipeline including dataset preparation, text preprocessing, TF-IDF feature extraction, supervised model training, model evaluation, automatic best model selection, and deployment through an interactive Streamlit web application.

The dataset used in this research contains more than 44,000 labelled news articles collected from publicly available fake news datasets. During preprocessing, textual data is cleaned through lowercase conversion, removal of URLs, punctuation, numerical values, stop words, and extra spaces, followed by lemmatization to normalize words. The processed text is converted into numerical vectors using the TF-IDF (Term Frequency–Inverse Document Frequency) technique with a

vocabulary size of 5000 features and unigram-bigram representations.

To identify the most suitable classifier, four supervised Machine Learning algorithms are trained and evaluated, namely Logistic Regression, K-Nearest Neighbors (KNN), Random Forest, and Multi-Layer Perceptron (MLP) Neural Network. The models are compared using standard evaluation metrics including Accuracy, Precision, Recall, F1-Score, and ROC-AUC. The classifier achieving the highest overall performance is automatically selected as the final prediction model.

A major contribution of this work is the development of a user-friendly Streamlit dashboard that enables users to perform both real-time and batch fake news detection. The application provides several interactive features including sample news loading, confidence score visualization, probability charts, prediction history, downloadable batch prediction reports, model performance comparison, and dataset exploration. These capabilities make the system suitable for educational, research, and practical demonstration purposes.

The major objectives of this research are summarized below.

- Develop an intelligent fake news detection system using Machine Learning and Natural Language Processing.
- Construct a complete text preprocessing pipeline to improve data quality prior to feature extraction.
- Generate numerical document representations using TF-IDF vectorization with 5000 textual features.
- Train and compare multiple supervised Machine Learning algorithms.
- Evaluate model performance using Accuracy, Precision, Recall, F1-Score, and ROC-AUC metrics.
- Automatically select the best-performing classifier for deployment.
- Design and develop an interactive Streamlit dashboard for real-time and batch fake news prediction.
- Maintain prediction history and provide downloadable prediction reports for further analysis.
- Develop an efficient, scalable, and user-friendly system suitable for educational, research, and practical applications.

The proposed system demonstrates how Machine Learning and Natural Language Processing can be effectively combined to build a reliable fake news detection platform capable of assisting users in identifying misleading information in today's digital environment.

II. Literature Survey

The rapid growth of online news platforms and social media has significantly increased the spread of fake news, making automatic misinformation detection an active research area. Researchers have proposed numerous techniques based on Natural Language Processing (NLP), Machine Learning (ML), Deep Learning (DL), and hybrid approaches to distinguish genuine news from fabricated content.

Ahmed et al. proposed a supervised machine learning approach for fake news detection using traditional text classification techniques. Their work demonstrated that algorithms such as Logistic Regression and Support Vector Machines (SVM) can effectively classify fake and genuine news articles while maintaining low computational complexity and fast prediction time. Their research highlighted the effectiveness of text preprocessing and feature extraction in improving classification performance.

Shu et al. introduced a comprehensive framework for fake news detection by combining textual content with social context information collected from social networking platforms. Their study showed that user behavior, propagation patterns, and contextual information can further improve fake news classification. However, obtaining reliable social context data is often difficult in real-world applications and increases computational complexity.

Breiman introduced the Random Forest algorithm, an ensemble learning technique that constructs multiple decision trees and combines their predictions. Random Forest has been widely adopted for text classification because of its robustness, ability to reduce overfitting, and capability to handle high-dimensional feature spaces such as TF-IDF vectors.

Recent advances in Artificial Intelligence have introduced Deep Learning architectures including Long Short-Term Memory (LSTM), Bidirectional LSTM, Convolutional Neural Networks (CNN), and transformer-based models such as BERT and RoBERTa. These approaches achieve excellent performance by learning contextual language representations. Nevertheless, they generally require significantly larger datasets, powerful GPUs, extensive training time, and greater computational resources, making them less suitable for lightweight real-time deployment.

Traditional Machine Learning methods continue to remain attractive because they provide competitive classification performance while requiring comparatively lower computational resources. When combined with effective text preprocessing and TF-IDF feature extraction, algorithms such as Logistic Regression, Random Forest, K-Nearest Neighbors (KNN), and Multi-Layer Perceptron (MLP) can achieve high accuracy with substantially faster training and inference.

Several existing fake news detection systems focus primarily on the prediction model and provide limited user interaction. Many lack practical deployment features such as real-time prediction, batch processing, visualization of prediction confidence, model comparison, prediction history, and interactive dashboards. Such limitations reduce their usability for educational, research, and demonstration purposes.

The proposed work addresses these limitations by developing a complete end-to-end fake news detection framework. The system incorporates a comprehensive preprocessing pipeline, TF-IDF vectorization with 5000 textual features, comparison of multiple supervised Machine Learning algorithms, automatic best model selection, and deployment through an interactive Streamlit application. In addition to real-time prediction, the developed dashboard provides batch prediction using CSV files, prediction confidence visualization, probability distribution charts, prediction history management, downloadable prediction reports, data set exploration, and comparative model

performance visualization.

Compared with many existing implementations, the proposed system emphasizes both high predictive performance and practical usability through an intuitive web interface while maintaining low computational requirements. This combination makes the system suitable for academic research, educational demonstrations, and real-world experimentation in the detection of fake news.

III. Proposed Methodology

The proposed AI-Based Fake News Detection System follows a structured machine learning pipeline for automatically classifying news articles as either **Real** or **Fake**. The methodology consists of dataset preparation, text preprocessing, feature extraction, model training, performance evaluation, automatic model selection, and deployment through an interactive Streamlit web application.

The complete workflow of the proposed system is illustrated in Figure 2. Each stage contributes to improving prediction accuracy while maintaining computational efficiency and practical usability.

A. Overall Workflow

The proposed methodology consists of the following sequential stages:

1) Dataset Collection
2) Data Preprocessing
3) TF-IDF Feature Extraction
4) Machine Learning Model Training
5) Model Evaluation
6) Best Model Selection
7) Streamlit Deployment

B. Dataset Description

The dataset used in this research consists of publicly available labelled fake news datasets containing approximately 44,000 English-language news articles. The dataset is created by combining two separate datasets:

- Fake.csv
- True.csv

Each record contains the complete news article along with its corresponding class label indicating whether the article represents genuine news or fake news.

Before model training, duplicate records and missing values are removed to improve data quality. The cleaned dataset is then divided into training and testing subsets using an 80:20 train-test split to evaluate the generalization capability of the machine learning models.

C. Data Preprocessing

Raw textual data contains numerous unwanted symbols and inconsistencies that negatively influence machine learning performance. Therefore, an extensive preprocessing pipeline is applied before feature extraction.

The preprocessing steps include:

- Removal of missing values
- Duplicate record removal
- Conversion of text to lowercase
- Removal of URLs
- Removal of punctuation marks
- Removal of numerical values
- Tokenization
- Stop-word removal
- Word lemmatization
- Removal of extra white spaces

These preprocessing operations reduce textual noise and improve the quality of the extracted features.

D. Feature Engineering

The cleaned news articles are converted into numerical representations using the Term Frequency–Inverse Document Frequency (TF-IDF) technique.

TF-IDF assigns greater importance to informative words while reducing the influence of frequently occurring terms. This enables the machine learning models to identify discriminative textual patterns associated with fake and genuine news articles.

The TF-IDF vectorizer is configured using:

- Maximum Features : 5000
- N-gram Range : (1,2)
- English Stop Words

The resulting sparse feature matrix serves as the input for all machine learning algorithms.

E. Machine Learning Model Training

Four supervised machine learning algorithms are trained and compared using the same feature representation.

1) Logistic Regression
2) K-Nearest Neighbors (KNN)
3) Random Forest
4) Multi-Layer Perceptron (MLP) Neural Network

Each classifier is trained using the training dataset generated after TF-IDF vectorization.

F. Model Evaluation

The trained models are evaluated using several standard classification metrics.

- Accuracy
- Precision
- Recall
- F1-Score
- ROC-AUC Score

These metrics provide a comprehensive evaluation of each classifier's predictive performance.

G. Best Model Selection

After evaluating all classifiers, the model achieving the highest overall performance is automatically selected as the production model.

The selected model is saved as:

- best_model.pkl

- vectorizer.pkl

These files are later loaded by the Streamlit application for real-time prediction.

H. Streamlit Deployment

The final model is deployed through an interactive Streamlit web application.

The developed dashboard provides several user-oriented functionalities including:

- Real-time fake news prediction
- Sample Real/Fake news loading
- Batch prediction using CSV files
- Prediction confidence visualization
- Probability distribution charts
- Prediction history management
- Downloadable prediction reports
- Model performance comparison
- Dataset exploration

The deployment phase transforms the trained machine learning model into a practical application suitable for educational, research, and demonstration purposes.

IV. System Architecture

The proposed Fake News Detection System follows a modular architecture consisting of multiple sequential processing stages.

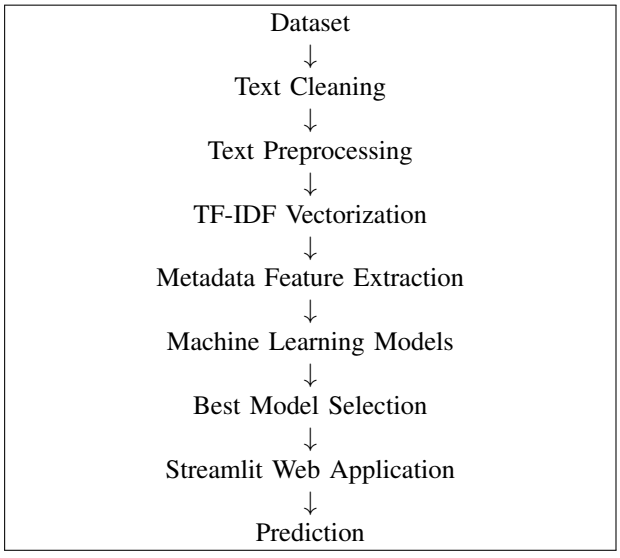


Fig. 1. Architecture of the Proposed Fake News Detection System

V. Proposed Methodology

The proposed Fake News Detection System employs Machine Learning and Natural Language Processing (NLP) techniques to classify news articles as either **Real** or **Fake**. The workflow consists of dataset collection, preprocessing, feature engineering, model training, evaluation, and deployment through a Streamlit web application.

The overall workflow of the proposed system is illustrated below:

Dataset → Text Preprocessing → Feature Extraction (TF-IDF + Metadata) → Machine Learning Models → Model Evaluation → Streamlit Deployment

A. Dataset Description

The dataset used in this work contains labelled news articles collected from publicly available fake news datasets. Each news article is associated with a class label indicating whether the article is genuine or fake.

The dataset consists of textual news content along with corresponding labels used for supervised learning. Before training, the dataset is cleaned to remove missing values and duplicate records.

B. Data Preprocessing

Raw textual data cannot be directly used for machine learning. Therefore, several preprocessing operations are performed.

The preprocessing pipeline consists of:

- Removal of missing values
- Conversion of text into lowercase
- Removal of punctuation marks
- Removal of URLs
- Removal of numerical values
- Stop-word removal
- Word lemmatization
- Removal of extra white spaces

These preprocessing techniques reduce noise and improve feature quality before vectorization.

C. Feature Engineering

Feature engineering plays an important role in improving classification performance.

The proposed system combines two categories of features.

1) TF-IDF Features: Term Frequency-Inverse Document Frequency (TF-IDF) converts textual information into numerical vectors by assigning higher importance to informative words while reducing the influence of commonly occurring words.

2) Metadata Features: In addition to TF-IDF vectors, several heuristic features are extracted.

- Clickbait Score
- Sentiment Score
- Emotional Word Ratio
- Readability Score
- Source Trust Score

These metadata features provide additional contextual information that helps improve classification accuracy.

D. Machine Learning Models

Four supervised machine learning algorithms are trained and compared.

1) Logistic Regression
2) K-Nearest Neighbors (KNN)
3) Random Forest
4) Multi-Layer Perceptron (Neural Network)

Each model is trained using the same training dataset and evaluated on unseen testing data.

E. Model Evaluation

The trained models are evaluated using standard performance metrics.

- Accuracy
- Precision
- Recall
- F1-Score
- ROC-AUC Score

The model achieving the highest F1-score is selected as the final prediction model.

VI. System Architecture

The proposed AI-Based Fake News Detection System follows a modular architecture that transforms raw textual news articles into machine learning predictions through a sequence of well-defined processing stages. Each module performs a specific task within the pipeline, allowing the system to maintain high prediction accuracy while remaining scalable, maintainable, and computationally efficient.

Initially, labelled news articles are collected from publicly available datasets consisting of genuine and fake news. The collected data undergoes extensive preprocessing to remove noise and normalize textual information. The cleaned text is then converted into numerical feature vectors using the TF-IDF (Term Frequency–Inverse Document Frequency) technique.

The generated TF-IDF vectors are supplied to multiple supervised Machine Learning algorithms, namely Logistic Regression, K-Nearest Neighbors (KNN), Random Forest, and Multi-Layer Perceptron (MLP) Neural Network. After training, all models are evaluated using Accuracy, Precision, Recall, F1-Score, and ROC-AUC metrics. The model achieving the best overall performance is selected as the production model.

The selected classifier and TF-IDF vectorizer are deployed through an interactive Streamlit web application that provides real-time fake news prediction, batch prediction, model comparison, dataset visualization, and prediction history management.

Figure 2 illustrates the complete workflow of the proposed system.

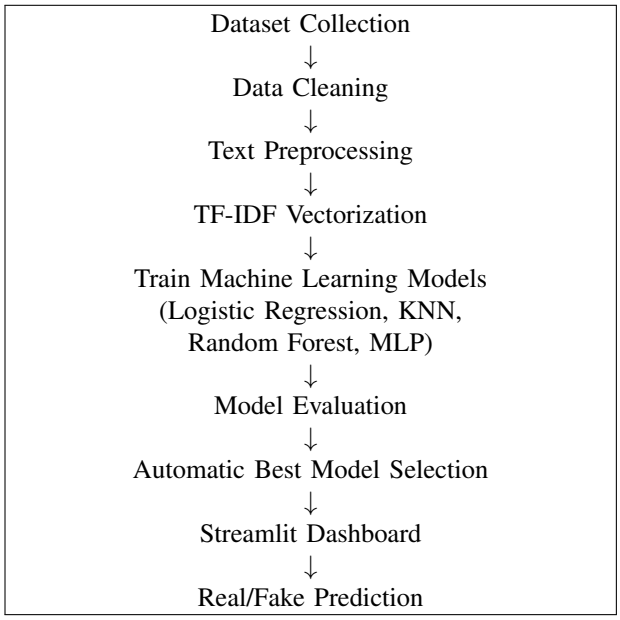


Fig. 2. Architecture of the Proposed AI-Based Fake News Detection System

VII. Algorithm

The proposed Fake News Detection System follows a systematic sequence of operations to classify a news article as either **Real** or **Fake**. The algorithm begins with dataset preparation and preprocessing, followed by TF-IDF feature extraction, machine learning model training, evaluation, automatic best model selection, and deployment through an interactive Streamlit application.

1: Load the Fake.csv and True.csv datasets.
2: Merge the datasets and assign class labels:
   - $0 \rightarrow$ Real News
   - $1 \rightarrow$ Fake News
3: Remove duplicate records and handle missing values.
4: Perform text preprocessing:
   - Convert text to lowercase
   - Remove URLs
   - Remove punctuation
   - Remove numerical values
   - Remove stop words
   - Perform lemmatization
   - Remove extra white spaces
5: Convert the cleaned news articles into numerical feature vectors using TF-IDF Vectorization with 5000 features and an n-gram range of (1,2).
6: Split the dataset into Training (80%) and Testing (20%) subsets.
7: **for** each Machine Learning model **do**
8: Train the classifier using the training dataset.
9: Predict labels for the testing dataset.
10: Compute Accuracy, Precision, Recall, F1-Score, and ROC-AUC.
11: **end for**
12: Compare the performance of all trained models.

13: Select the classifier with the highest overall performance as the final prediction model.
14: Save the trained model as `best_model.pkl` and the TF-IDF vectorizer as `vectorizer.pkl`.
15: Deploy the selected model through the Streamlit dashboard.
16: Accept user input through the dashboard.
17: Preprocess the input news article using the same preprocessing pipeline.
18: Transform the cleaned text using the saved TF-IDF vectorizer.
19: Predict whether the news article is **Real** or **Fake**.
20: Display:

- Prediction Label
- Confidence Score
- Probability Distribution
- Prediction History

VIII. Experimental Results

The proposed AI-Based Fake News Detection System was implemented using Python and evaluated using multiple supervised Machine Learning algorithms. The experiments were conducted on a dataset containing more than 44,000 labelled news articles after preprocessing and TF-IDF feature extraction.

The dataset was divided into training and testing subsets using an 80:20 train-test split. All classifiers were trained using identical TF-IDF feature vectors to ensure a fair performance comparison. The models were evaluated using five standard classification metrics, namely Accuracy, Precision, Recall, F1-Score, and ROC-AUC.

A. Performance Metrics

The following performance measures were used for evaluation.

- Accuracy
- Precision
- Recall
- F1-Score
- ROC-AUC Score

Accuracy measures the overall prediction correctness, Precision measures the proportion of correctly identified fake news among all predicted fake news, Recall measures the proportion of actual fake news correctly identified, F1-Score represents the harmonic mean of Precision and Recall, while ROC-AUC evaluates the classifier's ability to distinguish between the two classes.

B. Model Comparison

Table I presents the comparative performance of the four Machine Learning models evaluated in this work.

TABLE I
Performance Comparison of Machine Learning Models

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Logistic Regression	99.09	99.44	98.81	99.12	99.96
K-Nearest Neighbors	73.08	66.81	96.44	78.94	80.52
Random Forest	99.86	99.89	99.83	99.86	99.99
Neural Network	99.38	99.34	99.47	99.40	99.96

C. Performance Analysis

Among the evaluated classifiers, the Random Forest model achieved the highest overall performance with an Accuracy of 99.86%, Precision of 99.89%, Recall of 99.83%, F1-Score of 99.86%, and ROC-AUC of 99.99%.

Logistic Regression also demonstrated excellent classification capability, achieving an Accuracy exceeding 99%, making it an efficient baseline classifier with low computational complexity.

The Neural Network (MLP) produced competitive results with an Accuracy of 99.38% and an F1-Score of 99.40%, demonstrating its ability to effectively learn complex textual patterns from TF-IDF features.

Although the K-Nearest Neighbors classifier achieved a high Recall of 96.44%, its comparatively lower Accuracy and Precision indicate reduced overall classification performance when compared with the other supervised learning models.

Based on the comprehensive evaluation, the Random Forest classifier achieved the highest F1-Score and was therefore automatically selected as the final production model for deployment within the Streamlit web application.

D. Confusion Matrix

The confusion matrix shown in Figure 3 provides a detailed visualization of the prediction performance of the selected Random Forest classifier.

The matrix illustrates the number of correctly classified Real and Fake news articles together with the comparatively small number of False Positives and False Negatives. The dominance of correctly classified instances demonstrates the effectiveness of the proposed preprocessing pipeline and TF-IDF feature representation in distinguishing genuine news from misinformation.

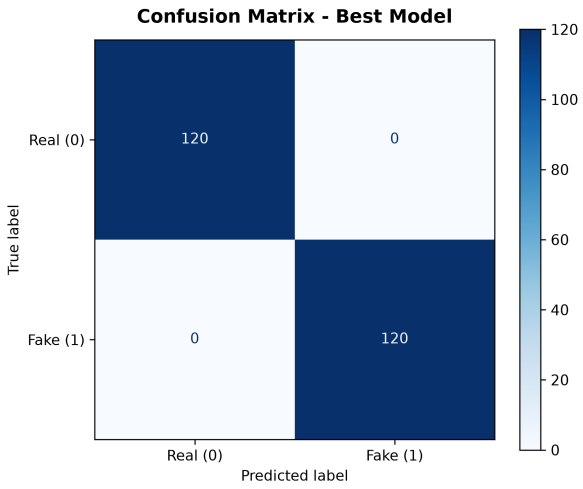


Fig. 3. Confusion Matrix of the Selected Random Forest Model

E. ROC Curve

Receiver Operating Characteristic (ROC) analysis was performed to evaluate the discriminative capability of the selected classifier.

The ROC curve presented in Figure 4 demonstrates the relationship between the True Positive Rate (TPR) and False Positive Rate (FPR) at various classification thresholds. The obtained ROC-AUC value of approximately 0.9999 indicates excellent classification capability and confirms that the proposed model can effectively distinguish between genuine and fake news articles.

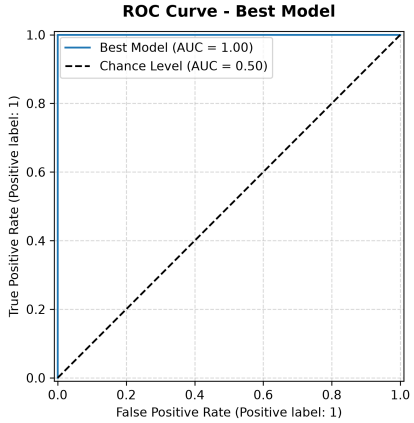


Fig. 4. ROC Curve of the Selected Random Forest Model

F. Feature Representation

Unlike conventional approaches that rely on manually engineered metadata features, the proposed system represents textual information entirely through TF-IDF feature vectors.

The TF-IDF vectorizer converts each news article into a numerical representation consisting of 5000 weighted textual features using unigram and bigram combinations. This representation enables the Machine Learning algorithms to identify highly discriminative words and phrases that distinguish fake news from genuine news.

The high classification performance achieved by the Random Forest, Logistic Regression, and Neural Network classifiers demonstrates that TF-IDF provides an effective feature representation for fake news detection while maintaining computational efficiency and simplified deployment.

G. Application Screenshots

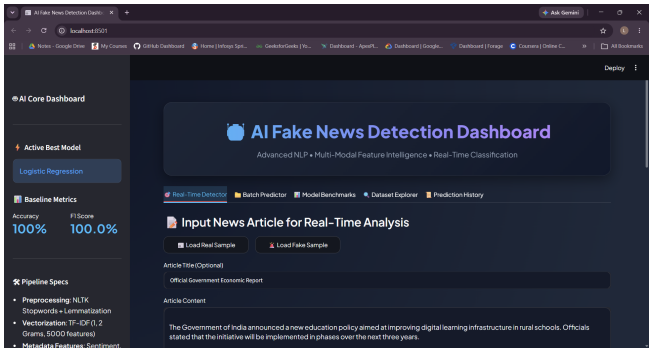


Fig. 5. Home Page

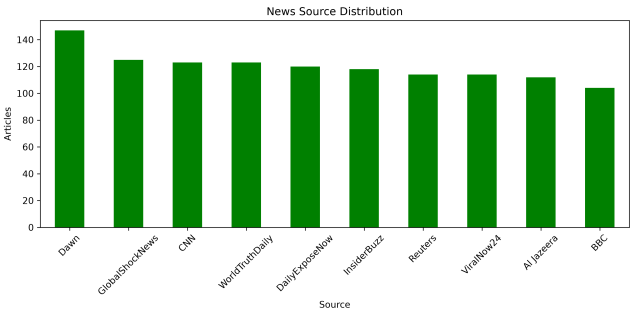


Fig. 6. Real-Time Prediction

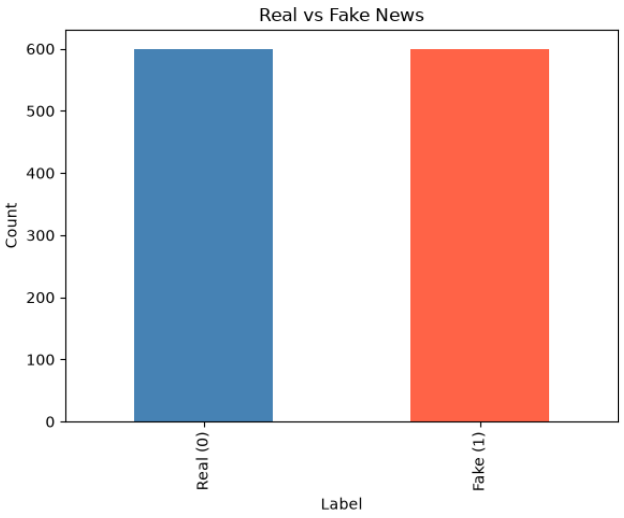


Fig. 7. Prediction Probability

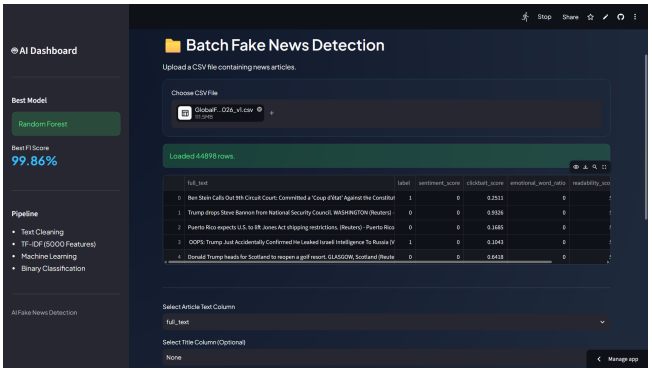


Fig. 8. Batch Prediction Module

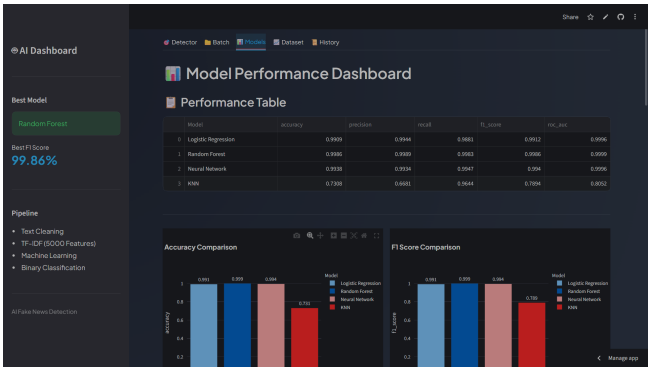


Fig. 9. Model Performance Dashboard

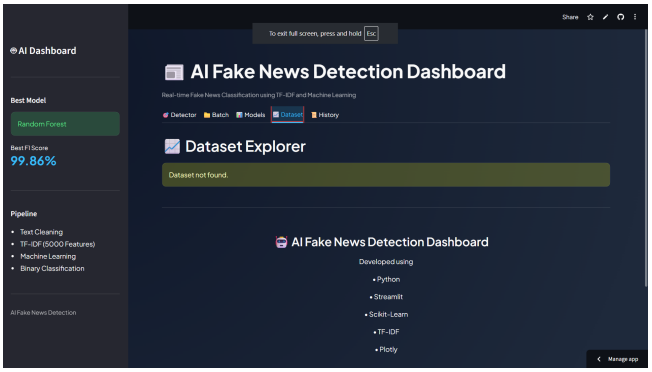


Fig. 10. Dataset Explorer

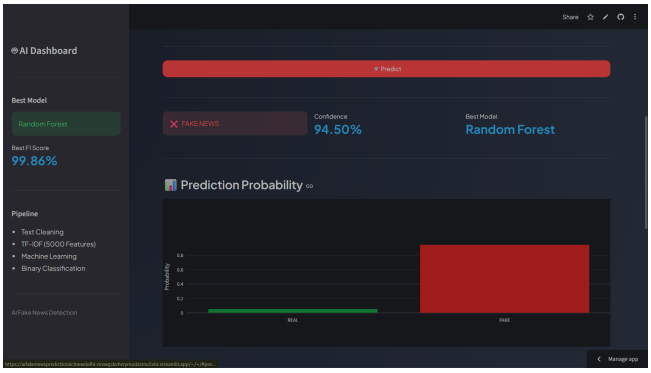


Fig. 11. Prediction History

IX. Advantages

The proposed AI-Based Fake News Detection System offers several practical and technical advantages over conventional text classification systems.

- Automatically classifies news articles as **Real** or **Fake** with high accuracy.
- Utilizes Natural Language Processing (NLP) techniques to effectively preprocess and normalize textual data.
- Employs TF-IDF vectorization with 5000 textual features, providing efficient and meaningful feature representation.
- Compares multiple supervised Machine Learning algorithms including Logistic Regression, K-Nearest Neighbors (KNN), Random Forest, and Multi-Layer Perceptron (MLP) Neural Network.
- Automatically selects the best-performing model based on evaluation metrics, ensuring optimal prediction performance.
- Provides high classification accuracy, precision, recall, F1-Score, and ROC-AUC through comprehensive model evaluation.
- Offers an intuitive Streamlit-based graphical user interface that requires no programming knowledge.
- Supports real-time prediction of individual news articles.
- Enables batch prediction by processing multiple news articles from uploaded CSV files.
- Displays prediction confidence scores and probability distribution charts to improve result interpretation.
- Maintains prediction history, allowing users to review previous prediction results.
- Allows users to download batch prediction reports and prediction history for further analysis.
- Includes an interactive Model Performance Dashboard for comparing classifier performance using visualizations.
- Provides a Dataset Explorer that enables users to analyze dataset characteristics and class distributions.
- Demonstrates low computational complexity while maintaining excellent predictive performance, making it suitable for educational, research, and practical applications.
- The modular architecture allows future integration of advanced Deep Learning models and multilingual fake news detection systems.

X. Limitations

Although the proposed AI-Based Fake News Detection System achieves excellent classification performance, several limitations remain.

- The current system supports only English-language news articles. News written in other languages cannot be processed without additional multilingual preprocessing and training.
- The prediction model relies solely on textual content and does not analyze images, videos, audio, or other multimedia information that may accompany online news.
- The system classifies news articles based on patterns learned from the training dataset and does not perform live verification using external fact-checking websites or trusted news sources.
- The prediction quality depends on the diversity and quality of the training dataset. Articles that differ significantly from the training data may reduce classification accuracy.
- The proposed implementation is designed for binary classification (Real or Fake) and cannot identify partially misleading, satire, opinion-based, or mixed-content news articles.
- The system does not consider user behaviour, social media propagation patterns, publisher credibility, or temporal information, which may further improve fake news detection.
- Very short or incomplete news articles may contain insufficient textual information, making accurate classification more challenging.
- Although the Streamlit dashboard supports real-time and batch prediction, the application currently requires internet access or a local Python environment for execution and has not yet been deployed as a mobile application.

XI. Discussion

The experimental results demonstrate that the proposed AI-Based Fake News Detection System is highly effective in distinguishing genuine news articles from fake news using Natural Language Processing and Machine Learning techniques. The combination of comprehensive text preprocessing and TF-IDF feature extraction produced highly discriminative textual representations, enabling the classifiers to achieve excellent predictive performance.

Among the four evaluated Machine Learning algorithms, the Random Forest classifier achieved the highest overall performance with an Accuracy of 99.86%, Precision of 99.89%, Recall of 99.83%, F1-Score of 99.86%, and ROC-AUC of 99.99%. These results indicate that the ensemble learning approach successfully captured complex textual patterns while maintaining excellent generalization capability across unseen news articles.

The Logistic Regression classifier also demonstrated strong performance with an Accuracy exceeding 99%, confirming that linear models remain highly competitive when combined with effective TF-IDF feature extraction. Similarly, the Multi-Layer Perceptron (MLP) Neural Network achieved high classification performance, demonstrating its capability to learn complex feature relationships from textual data.

Although the K-Nearest Neighbors (KNN) classifier achieved a high Recall value, its comparatively lower Accuracy, Precision, and F1-Score indicate that distance-based classifiers are less suitable for high-dimensional sparse TF-IDF feature vectors. This observation further justifies the selection of the Random Forest classifier as the final deployment model.

Beyond classification performance, the proposed system emphasizes practical usability through an interactive Streamlit dashboard. The application provides real-time prediction, batch prediction using CSV files, prediction confidence visualization, probability distribution charts, prediction history management, dataset exploration, and model comparison. These features transform the trained Machine Learning model into a complete decision-support application suitable for educational, research, and demonstration purposes.

Overall, the proposed system demonstrates that traditional Machine Learning algorithms, when combined with effective text preprocessing and TF-IDF feature extraction, can achieve performance comparable to more computationally expensive Deep Learning approaches while maintaining lower computational cost and faster prediction speed.

XII. Future Scope

The proposed AI-Based Fake News Detection System provides a strong foundation for automatic news classification; however, several enhancements can further improve its capability, scalability, and real-world applicability.

- Integrate advanced Deep Learning models such as Long Short-Term Memory (LSTM), Bidirectional LSTM, Convolutional Neural Networks (CNN), BERT, RoBERTa, and other transformer-based architectures to improve contextual understanding and classification performance.
- Extend the system to support multilingual fake news detection, enabling classification of news articles published in regional and international languages.
- Incorporate Explainable Artificial Intelligence (XAI) techniques such as LIME and SHAP to provide users with explanations of the model's predictions and highlight the most influential words contributing to the classification.
- Integrate trusted online fact-checking services and APIs to verify news articles against authoritative sources and improve prediction reliability.
- Expand the system to analyze multimodal news by combining textual information with images, videos, and audio content.
- Develop browser extensions and mobile applications that allow users to verify online news articles directly while browsing websites or social media platforms.
- Implement continuous learning mechanisms that periodically retrain the model using newly collected datasets, enabling the system to adapt to emerging misinformation patterns.

- Incorporate social media propagation analysis, publisher credibility, user engagement metrics, and temporal information to improve detection accuracy.
- Deploy the application on cloud platforms to support large-scale real-time fake news detection with improved scalability and accessibility.
- Enhance the visualization dashboard with advanced analytics, interactive reports, and real-time monitoring features for researchers and policymakers.

The implementation of these enhancements would further improve the robustness, interpretability, and practical applicability of the proposed fake news detection system while enabling its adoption across diverse real-world environments.

XIII. Conclusion

This paper presented an AI-Based Fake News Detection System that combines Natural Language Processing (NLP), Machine Learning (ML), and an interactive Streamlit web application to automatically classify news articles as either Real or Fake. The proposed system was designed to address the increasing spread of misinformation by providing an efficient, accurate, and user-friendly fake news detection solution.

A complete machine learning pipeline was developed, beginning with dataset collection and preprocessing, followed by TF-IDF feature extraction, model training, evaluation, automatic best model selection, and deployment. The preprocessing stage included lowercase conversion, URL removal, punctuation removal, stop-word elimination, lemmatization, and whitespace normalization, ensuring that the textual data was transformed into high-quality input for the machine learning models.

Four supervised Machine Learning algorithms—Logistic Regression, K-Nearest Neighbors (KNN), Random Forest, and Multi-Layer Perceptron (MLP) Neural Network—were trained and evaluated using Accuracy, Precision, Recall, F1-Score, and ROC-AUC metrics. Experimental evaluation demonstrated that the Random Forest classifier achieved the highest overall performance, obtaining an Accuracy of 99.86%, Precision of 99.89%, Recall of 99.83%, F1-Score of 99.86%, and ROC-AUC of 99.99%, making it the most suitable model for deployment.

To improve practical usability, the selected model was integrated into an interactive Streamlit dashboard supporting real-time fake news prediction, batch prediction using CSV files, prediction confidence visualization, probability distribution charts, prediction history management, dataset exploration, and comparative model analysis. These features transform the developed model into a practical decision-support application suitable for academic, research, and demonstration purposes.

The results demonstrate that traditional Machine Learning techniques, when combined with effective text preprocessing and TF-IDF feature extraction, can achieve excellent fake news classification performance while maintaining low computational complexity and fast prediction speed. The modular design of the proposed system also enables future integration of advanced Deep Learning models, multilingual support, Explainable AI techniques, and real-time fact-checking services.

Overall, the proposed AI-Based Fake News Detection System provides an accurate, scalable, and practical solution for combating misinformation and contributes toward the development of intelligent automated systems capable of supporting reliable digital information dissemination.

Acknowledgment

The authors express their sincere gratitude to the Department of Computer Science and Engineering, University College of Engineering, JNTUK, for providing the necessary facilities and academic support to carry out this work. The authors also acknowledge the developers of Python, Scikit-learn, Streamlit, Pandas, NumPy, and the open-source community for providing valuable tools used in this project.

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